NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CHEMISTRY

9~~64~~729/03

Paper 3 Free Response

~~21~~ 18 September 2018~~6~~

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2 hours

Additional Materials: Writing Paper
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, ~~highlighters,~~ glue or correction fluid.

Section A
Answer **all** questions.

Section B
Answer **one** question.

A Data Booklet is provided.
The use of an approved scientific calculator is expected, where appropriate.
~~Answer any **four** questions.~~
~~A Data Booklet is provided.~~
~~You are reminded of the need for good English and clear presentation in your answers.~~

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

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Section A

Answer **all** questions in this section.

1 Cobalt is a typical transition element which is commonly used as a catalyst and metal for electroplating. Cobalt also forms complex ions with ligands such as H_2O and NH_3 to give various coloured octahedral complexes such as $[Co(H_2O)_6]^{2+}$ and $[Co(NH_3)_6]^{2+}$ which are pink and yellow respectively.

(a) The ligand exchange in octahedral complexes is one of the most extensively studied reactions in transition metals.

An example of a ligand exchange reaction involving cobalt-(II) ions is:

$$[Co(H_2O)_6]^{2+} + 6NH_3 \rightleftharpoons [Co(NH_3)_6]^{2+} + 6H_2O$$

(i) Explain why cobalt forms coloured complexes. [3]

(ii) Suggest why $[Co(NH_3)_6]^{2+}$ is of a different colour from $[Co(H_2O)_6]^{2+}$. [1]

(iii) A student wishes to investigate the kinetics of the ligand exchange reaction of $[Co(H_2O)_6]^{2+}$ to form $[Co(NH_3)_6]^{2+}$ by using a spectrometer. This machine measures the amount of light that is absorbed when a specific wavelength of visible light is shone through a few cm^3 of the coloured solution. It does this by comparing the amount of light passing through the sample with the amount of light passing through the pure solvent.

The spectrometer is set to use the wavelength of light that is absorbed most strongly by the complex ion. The amount of light absorbed is expressed as an absorbance value. The more concentrated the solution, the higher the absorbance value. The temperature of the sample in the spectrometer can be thermostatically controlled for reaction rate analysis for which the sample has to be kept at a constant temperature.

Outline the experimental procedure on how the student would accurately determine the initial rate of the ligand exchange reaction at 5 °C.
~~[3]~~
The details of the use of ~~No details regarding use of~~ specific glassware for measurement are **not** required. [3]

(iv) When $[Co(H_2O)_6]^{2+}$ is mixed with an excess of NH_3 (aq), each H_2O molecule is replaced by a NH_3 molecule one at a time. Given that the stepwise formation of $[Co(NH_3)_6]^{2+}$ from $[Co(H_2O)_6]^{2+}$ undergoes a dissociative mechanism which resembles a S_N1 mechanism in organic chemistry.
Suggest a possible mechanism for the formation of $[Co(H_2O)_5NH_3]^{2+}$ from $[Co(H_2O)_6]^{2+}$ and show clearly how the shape of the complex ion changes.
In your mechanism, show appropriate curly arrows, lone pairs and dipoles.
[3]

(v) State the rate equation for the above ligand exchange reaction. [1]

(vi) Hence, predict and explain the effect on the rate of reaction, if any, when the ammonia ligand is replaced ~~by~~with a fluoride ion.
[1]

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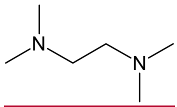
(b) $[Co(H_2O)_6]^{2+}$ can also undergo ligand exchange reactions with TMEDA to form $[Co(TMEDA)_3]^{2+}$.



TMEDA

By considering the entropy and enthalpy changes during the formation of $[Co(TMEDA)_3]^{2+}$ from $[Co(H_2O)_6]^{2+}$ and that of $[Co(NH_3)_6]^{2+}$ from $[Co(H_2O)_6]^{2+}$, suggest how the standard Gibbs free energy change of the two reactions will compare in sign and in magnitude.

Hence, predict which reaction will be ~~the~~ more spontaneous. Explain your reasoning.
[3]

(c) **(i)** Draw a fully labelled diagram of an electrochemical cell composed of a standard $Cl_2|Cl^-$ electrode and a standard $Co^{2+}|Co$ electrode. Indicate the direction of the electron flow. [3]

(ii) Calculate the $E^\ominus_{cell}$ of the electrochemical cell and write a balanced equation for the cell reaction.
[~~1~~2]

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(iii) Using your answer in **(ii)**, calculate ΔG for the cell reaction. [1]

(iv) Use the *Data Booklet* to suggest the effect on the cell potential of this cell of adding excess aqueous ammonia to the $Co^{2+}|Co$ half cell. Explain your answer. [1]

[Total: 2~~1~~2]

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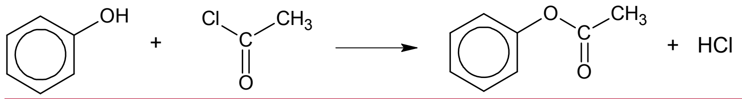
~~**Answer any four questions.**~~

~~1~~2**(a)** Aspirin is one of the most widely used drug in the world. It is a powerful analgesic (pain reliever), antipyretic (fever reducer) and anti-inflammatory drug.

It is synthesised using 2-hydroxybenzoic acid and ethanoic anhydride. 8 – 10 drops of 85% phosphoric acid which catalyses the reaction is added. The reaction mixture is then heated under reflux for around fifteen minutes. The other product of this reaction is ethanoic acid.

2-hydroxybenzoic acid ethanoic anhydride

Ethanoyl chloride and phenol can undergo condensation reaction.



Ethanoic anhydride and 2-hydroxybenzoic acid can undergo a similar reaction to form aspirin.

(i) Draw the structure of aspirin. [1]

(ii) Draw a labelled diagram of the assembled apparatus for the synthesis of aspirin. [3]

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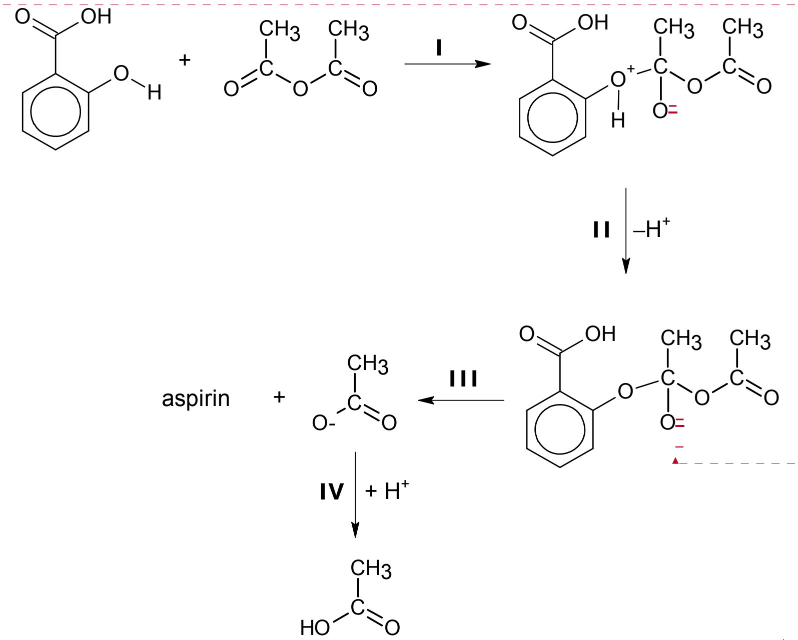
The mechanism for the reaction between 2-hydroxybenzoic acid and ethanoic anhydride involves four steps. It is proposed as below:



(iii) Using the information given above, state the type of mechanism in step **I**. [1]

(iv) Copy and complete the whole mechanism above by showing any relevant charges, lone pairs of electrons and movement of electrons in your answer. [3]

(v) State a reason why ethanoic anhydride is used rather than ethanoyl chloride for the synthesis of aspirin. [1]

(b) In 1911, the French chemist F.A.V. Grignard reacted small pieces of magnesium with a warm solution of bromoethane in a dry, non-polar solvent and obtained a solution containing ethylmagnesium bromide, C_2H_5MgBr. Many Grignard reagents, with different alkyl or aryl groups, have now been prepared and are widely used in organic syntheses. A typical example of the use of a Grignard reagent is the two-step reaction of C_2H_5MgBr with propanone, CH_3COCH_3, to form 2-methylbutan-2-ol.

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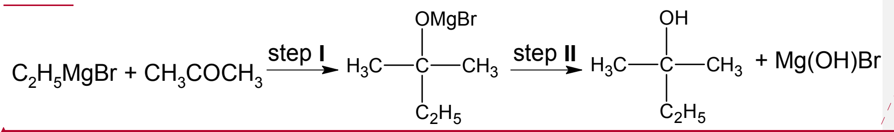
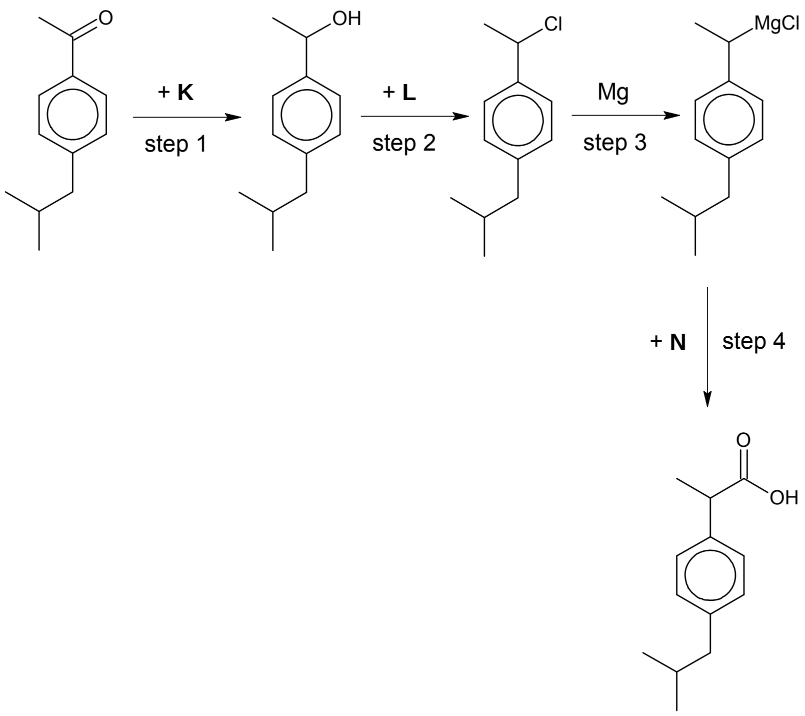


Suggest the type of reaction which occurs in step **II**. [1]

(c) The following scheme shows the synthesis of ibuprofen which is an alternative medication to aspirin. In step 4, the Grignard reagent readily converts into a carboxylic acid.



ibuprofen

(i) Suggest the identity of the reagent **K** in step 1. [1]

(ii) Suggest the identity of the reagent **L** in step 2. [1]

(iii) Suggest the identity of the reagent **N** in step 4. [1]

(d) Suggest a simple chemical test that could be used to distinguish between aspirin and ibuprofen. You should state what you would observe for **each** compound. [3]

[Total: 16]

~~Copper(I) sulfate, Cu_2SO_4, can be made from copper(I) oxide under non-aqueous conditions. On adding this salt to water, it immediately undergoes a disproportionation reaction.~~

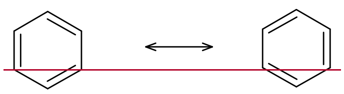
~~(i) Suggest, with a reason, the colour of copper(I) sulfate. [2]~~

~~(ii) Using suitable data from the *Data Booklet*, explain why the disproportionation reaction occurs, and write an equation for it. [3]~~

~~(b) Palladium(II) salts can form square planar complexes. Successive addition of ammonia and hydrogen chloride to an aqueous palladium(II) salt produces, under different conditions, three compounds with empirical formula $PdN_2H_6Cl_2$. Two of these, **A** and **B**, are non-ionic, with M_r = 211. **A** has a dipole moment, whereas **B** has none. The third compound, **C**, is ionic, having M_r = 422, and contains palladium in both its cation and anion.~~

~~For each **A**, **B** and **C**, deduce a structure that fits the above data, explaining your reasons fully. [6]~~

~~(c) Benzene ring is often represented as a structure that has a ring within the hexagon. Alternatively, chemists have also represented the structure of benzene in the following forms, known as resonance structures.~~

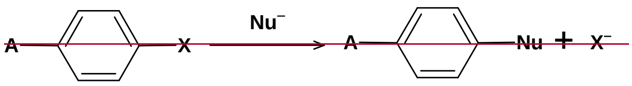


Benzene (two resonance forms)

~~The resonance relationship is indicated by the double headed arrow between them. The only difference between resonance forms is the placement of the pi electrons and non-bonding electrons.~~

~~Most aromatic compounds undergo electrophilic substitution. However aryl halides undergo a limited number of substitution reactions with strong nucleophiles.~~

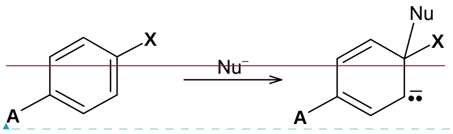
~~An example of a reaction is as follows:~~



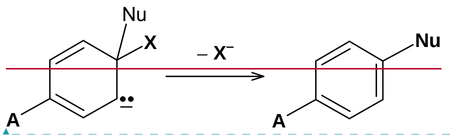
~~where **A** is an electron withdrawing group and **X** is a halogen~~

~~The mechanism of this reaction has two steps:~~
- ~~addition of the nucleophile~~
- ~~elimination of the halogen leaving group~~

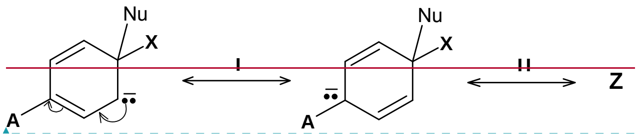
~~3 **Step 1** involves the addition of the nucleophile (Nu⁻). The Nu⁻ attacks the carbon atom bonded to a halogen, causing the pi bond to break. A resonance stabilised carbanion with a new C–Nu bond is formed. The aromatic ring is destroyed in this step.~~



~~**Step 2** involves the loss of the halogen **X**, reforming the aromatic ring.~~

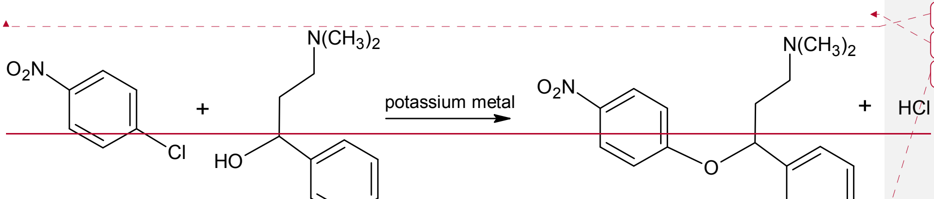


~~Two other resonance structures of the intermediate in **Step 1** are shown below:~~



~~**(i)** Copy the above diagram and draw the resonance structure, **Z**. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons in forming **Z**.~~ ~~[2]~~

~~The reaction below shows the synthesis of compound, **D**.~~

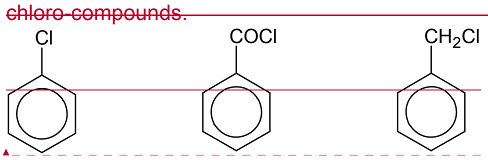


D

~~**(ii)** Suggest the role of potassium metal in the reaction.~~ ~~[1]~~

~~**(iii)** Use the information given above to draw out the full mechanism for the reaction that forms **D**, labelling the slow and fast steps. In your answer, showing any relevant charges, lone pair of electrons and movement of electrons.~~ ~~[3]~~

~~**(d)** Describe and explain the relative ease of hydrolysis of the following three chloro-compounds.~~ ~~[3]~~



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~~[Total: 20]~~

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~~**2** Sulfuric acid, H_2SO_4, can behave as an acid, an oxidising agent or as a dehydrating agent in various reactions.~~

~~**(a)** Draw a diagram to illustrate the shape of pure sulfuric acid and indicate the bond angle about the sulfur atom. [2]~~

~~**(b)** The Contact Process is used for the manufacture of sulfuric acid. One of the reactions that takes place is the following reversible reaction:~~

~~$2SO_2(g) + O_2(g)$ Ý $2SO_3(g)$ $\Delta H = -197$ kJ mol^{-1}~~

~~Sulfur dioxide and oxygen in a 2:1 molar ratio at a total initial pressure of 3 atm is passed over a catalyst in a fixed volume vessel at 400 °C. When equilibrium is established, the percentage of sulfur trioxide in the mixture of gases is found to be 30%.~~

~~**(i)** Write an expression for the equilibrium constant, K_p, of the reaction. [1]~~

~~**(ii)** Calculate the value of K_p at 400 °C, stating its units. [3]~~

~~**(iii)** How would the percentage conversion of SO_2 into SO_3 be affected when the pressure is raised? Explain. [2]~~

~~**(c)** Dilute sulfuric acid takes part in typical acid-base reactions and it can be used to distinguish the following solids: MgO, BaO and SiO_2.~~

~~State the observations, if any, to indicate the differences in their reaction when water is added to each solid followed by dilute sulfuric acid. [4]~~

~~**(d)** Sulfur dioxide is a major pollutant from sulfuric acid plants. The SO_2 emitted into the atmosphere is oxidised in the air, which then reacts with water to form sulfuric acid, hence causing acid rain:~~

~~$2SO_2(g) + O_2(g) + 2H_2O(l) \longrightarrow 2H_2SO_4(l)$ ΔH_1~~

~~Using the data below and data from **(b)**, construct an energy cycle to calculate~~

~~**(i)** the enthalpy change of formation of $SO_2(g)$, and hence~~

~~**(ii)** the enthalpy change of reaction, ΔH_1 for the above reaction.~~

~~Enthalpy change of formation of $H_2O(l)$ = –286 kJ mol^{-1}~~
~~Enthalpy change of formation of $H_2SO_4(l)$ = –811 kJ mol^{-1}~~
~~Enthalpy change of formation of $SO_3(g)$ = –493 kJ mol^{-1}~~
~~[4]~~

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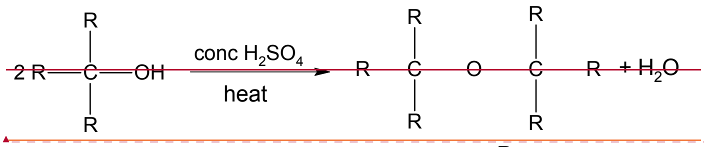
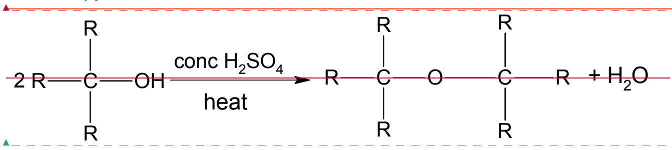
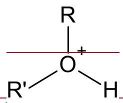
~~**(e)** Alcohols react with concentrated sulfuric acid at high temperatures to form alkenes. A common side reaction that can happen is the formation of ethers, which is also catalysed by concentrated sulfuric acid.~~





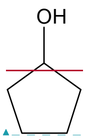
~~The mechanism occurs via 3 steps:~~

~~**Step 1:**~~
~~An acid base reaction in which H^+ from H_2SO_4 protonates the oxygen atom in alcohol. This step is very fast and reversible.~~

~~**Step 2:**~~
~~A second alcohol molecule functions as the nucleophile and attacks the product from **step 1**. The C–O bond is cleaved and a water molecule leaves the molecule. This creates an oxonium ion~~  ~~intermediate.~~

~~**Step 3:**~~
~~Another acid base reaction in which the proton in the oxonium ion is removed by a suitable base (in this case a water molecule) to give the ether product. This step is very fast and reversible.~~

~~**(i)** Draw the ether formed when cyclopentanol undergoes the above reaction.~~



~~cyclopentanol~~
~~[1]~~

~~**(ii)** Draw out the full mechanism for the reaction between two cyclopentanol molecules to form an ether. In your answer, show any relevant charges, lone pairs of electrons and movement of electrons. [3]~~

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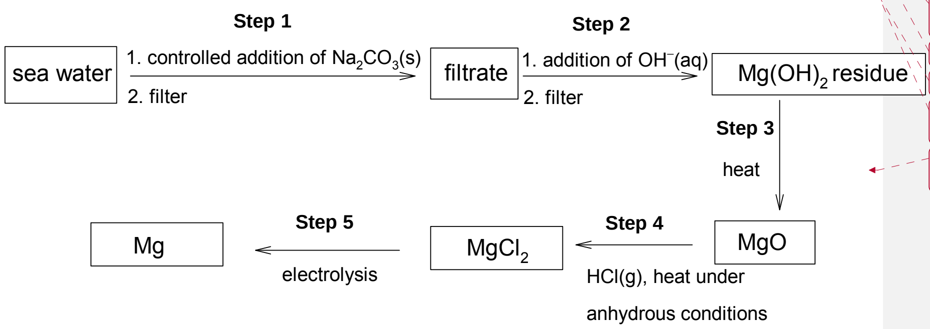
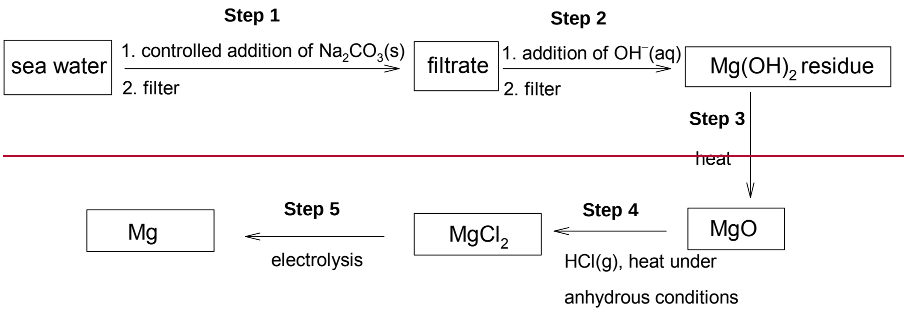
3 Magnesium is present as dissolved magnesium ions in sea water and is the only metal directly extracted from sea water. There is enough magnesium dissolved in the Earth's oceans to supply all of our magnesium needs for the next 1000 years.

~~Garlic contains many amino acids, minerals and enzymes. Garlic also contains at least 33 sulfur compounds like alliin, allicin and ajoene. The sulfur compounds are responsible for both garlic's pungent odour and many of its medical effects.~~

(a) Apart from magnesium ions, the two other most abundant cations found in sea water are sodium and calcium ions.

Magnesium can be extracted from sea water by the following steps:





Concentration of common ions in sea water:

ion	**concentration / mol dm^{-3}**
magnesium	0.056
calcium	0.010
sodium	0.457
chloride	0.535

The numerical values of solubility products are given below:

compound	**value of solubility product**
magnesium carbonate	1.00×10^{-5}
calcium carbonate	8.70×10^{-9}

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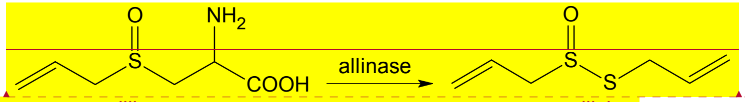
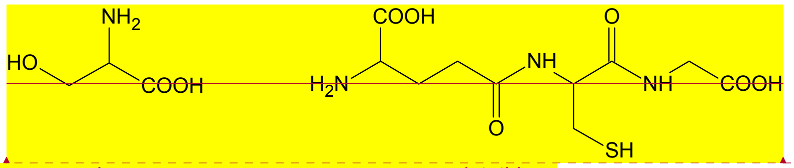
magnesium hydroxide	5.61×10^{-12}
calcium hydroxide	5.50×10^{-6}

~~When raw garlic is chopped or crushed, the enzyme allinase converts alliin into allicin, which is responsible for the odour of fresh garlic.~~



~~alliin~~ ~~allicin~~

~~Allicin can also be biosynthesised by using serine and glutathione.~~



~~serine~~ ~~glutathione~~

~~(i) Explain why raw chopped garlic has a stronger odour than when it is cooked. [1]~~

(i) Explain why the addition of sodium carbonate ~~ions~~ in **step 1** has to be controlled. [1]

(ii) Hence, state the cations present in the filtrate after **step 1** is carried out. [1]

(iii) What is the maximum mass of solid sodium carbonate that can be added to 1 dm^3 of sea water in **step 1**? [2]

(iv) Use the data provided to explain the following:

- Solid sodium carbonate was added to sea water (under controlled conditions) **before** the hydroxide ions.
- The reverse order (i.e. adding hydroxide ions before sodium carbonate) is **not** preferred over the extraction of magnesium.

~~(under controlled conditions)~~

~~sodium~~

[2]

(v) Calculate the minimum pH of the hydroxide solution required for precipitation of magnesium hydroxide in **step 2** if an equal volume of hydroxide ions was added to the filtrate. Give your answer to 2 decimal places. [2]

(b) (i) Write the equations that occur during the electrolysis of magnesium chloride in ~~S~~**step 5**. State clearly the reactions that occur at the cathode and the anode, and include state symbols. [2]

(ii) In a factory, a current of 95 kA was passed through a suitable setup for 24 hours. Assuming that the procedure is 90% efficient, calculate the mass of Mg that can be produced. [2]

(iii) Give a reason why electrolysis of magnesium chloride is preferred to that of magnesium oxide in this industrial process. [1]

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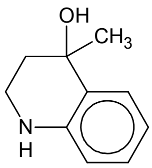
(c) ~~A is an organic compound.~~ When 0.46~~7~~8 g of an organic compound **A** was evaporated in a syringe, the volume of the vapour produced after correction to s.t.p was 60 cm^3. On heating with aqueous sodium hydroxide, **A** gives a compound that dissolves in water.

A reacts with aluminium oxide to give two products **B** and **C**. Both **B** and **C** react with HBr to give the same product **D**. **D** exhibits enantiomerism and exists as a pair of enantiomers.

A gives **E** when reacted with lithium aluminium hydride in dry ether.



E

(i) Prove that the molar~~ecular~~ mass of **A** is 177 g mol^{-1}. [1]

(ii) Hence, d~~D~~educe the structural formulae of all the above structures, and explain the chemistry involved. [6]

(iii) State the type of isomerism exhibited by **B** and **C.** Explain why **B** and **C** both give the same product **D** when reacted with HBr. [2]

[Total: 22]

~~**(ii)** When one molecule of serine reacts with one molecule of glutathione, it is possible to form two esters with different structural formulae.~~
~~Draw the structural formula of **each** of these esters. [2]~~

~~**(iii)** Draw the structural formulae of the products when glutathione is hydrolysed. [3]~~

~~Alliin has pK_a values of 1.84 and 8.45.~~

~~**(iv)** Make use of these pK_a values to suggest the major species present in solutions of alliin with the following pH values. [3]~~

- ~~pH 1~~
- ~~pH 7~~
- ~~pH 11~~

~~**v)** Calculate the pH of 0.10 mol dm^{-3} solution of alliin. [1]~~

~~**(vi)** With reference to the pK_a values, identify the major species formed when 10 cm^3 of 0.10 mol dm^{-3} NaOH is added to 10 cm^3 of 0.10 mol dm^{-3} protonated alliin. Hence, deduce whether the solution is acidic, neutral or alkaline. [2]~~

~~**(vii)** Sketch the pH-volume added curve you would expect to obtain when 30 cm^3 of 0.10 mol dm^{-3} NaOH is added to 10 cm^3 of 0.10 mol dm^{-3} protonated alliin.~~
~~Briefly describe how you have calculated the various key points on the curve. [4]~~

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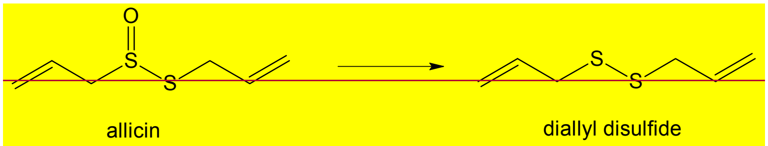
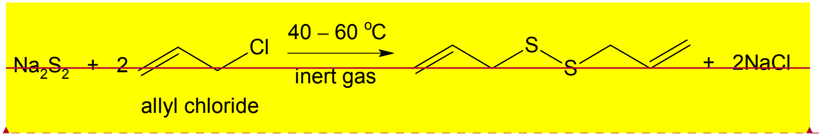
~~**(b)** Diallyl disulfide is one of the principal components of the distilled oil of garlic. It is a yellowish liquid which is insoluble in water and has a strong garlic odour. It is produced during the decomposition of allicin.~~



~~Diallyl disulfide can be produced industrially from sodium disulfide and allyl chloride at temperatures of 40 – 60 ¼C in an inert atmosphere.~~



~~**(i)** Give the *IUPAC* name of allyl chloride. [1]~~

~~**(ii)** Explain, in thermodynamic terms, suggest why diallyl disulfide is insoluble in water. [2]~~

~~**(iii)** State the type of reaction when diallyl disulfide is converted back to allicin. [1]~~

~~[Total: 20]~~

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Section B

Answer **one** question from this section.

4 Cycloalkanes are a homologous series of cyclic saturated hydrocarbons with the general formula C_nH_{2n} while n-alkanes are a homologous series of straight-chain saturated hydrocarbons with the general formula C_nH_{2n+2}.

n-alkanes	Boiling point / °C	Enthalpy change of combustion / kcal mol^{-1}	cycloalkanes	Boiling point / °C	Enthalpy change of combustion / kcal mol^{-1}
ethane	–89	–373.0	-	-	-
propane	–42	–530.4	cyclopropane	–33	–499.8
butane	–1	–687.8	cyclobutane	12	–656.0
pentane	36	–845.2	cyclopentane	49	–793.5
hexane	69	–1002.6	cyclohexane	81	–944.6
heptane	98	–1160.0	cycloheptane	119	–1108.3

(a) Explain the term "homologous series". [1]

(b) Explain the increase in magnitudes of both boiling point and enthalpy change of combustion from ethane to heptane. [3]

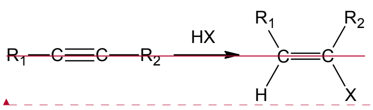
~~Alkynes are organic molecules which contain carbon-carbon triple bonds and are part of the homologous series with formula of~~ ~~C_nH_{2n-2}~~ ~~i.e.~~

$$R_1—C\equiv C—R_2$$

~~where R_1 and R_2 = H or alkyl or aryl groups~~

~~Alkynes exhibit similar chemical properties to alkenes.~~

~~e.g. addition reactions with electrophiles i.e. X_2 or HX to form alkenes~~



~~e.g. oxidation by hot concentrated $KMnO_4$ to form mixture of carboxylic acids~~

$$R_1—C\equiv C—R_2 \xrightarrow{KMnO_4} R_1COOH + R_2COOH$$

~~However, unlike alkenes, terminal alkynes are able to react with strong bases like sodium amide.~~

$$RC\equiv CH + NaNH_2 \longrightarrow RC\equiv C^-Na^+ + NH_3$$

~~**(a)** Ethyne, C_2H_2, is heated with excess sodium bromide and concentrated sulfuric acid to produce a dihalide, $C_2H_4Br_2$. The overall reaction may be considered to take place in two stages, the first between inorganic reagents only and the second involving the organic reagent.~~

~~**(i)** Write an equation for the first stage. [1]~~

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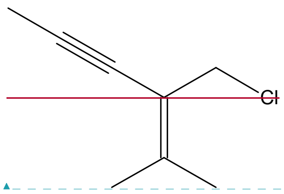
~~**(ii)** Suggest a structure for the dihalide formed. [1]~~

~~**(iii)** When the concentrated sulfuric acid is added to the reaction mixture, cooling is necessary to prevent the formation of inorganic by-products.~~
~~Write an equation to explain the formation of these inorganic by-products. [1]~~

~~**(b)** Compound **A**, is an enyne chloride (i.e. compounds that contains chloro, alkyne and alkene functional groups).~~



~~Compound **A**~~

~~One mole of compound **A** reacts with two moles of Br_2 to produce a mixture of 4 stereoisomers. Draw structures of the stereoisomers formed. [3]~~

(c) ~~Compound **B**, which is an isomer of Compound **A** and also an enyne chloride, is treated with sodium amide, $NaNH_2$ followed by heating under reflux to form compound **C**, C_8H_{10}. Compound **C** reacts with hot concentrated $KMnO_4$ to produce butane-1,4-dioic acid only.~~ Suggest why combustion tends to be incomplete as the alkane increases in molecular mass. [1]

Cyclopropane is a colourless gas with a "petroleum-like" odour. Unlike its straight-chain counterpart, it is considered to be highly strained and unstable. The instability of cyclic alkanes can be measured by calculating its "ring strain energy" using the formula below:

Ring strain energy =

$$\text{number of carbon atoms in cyclic alkane, A} \times \left(\left| \text{enthalpy change of combustion of a } CH_2 \text{ group in the cyclic alkane, A} \right| - \left| \text{enthalpy change of combustion of a } CH_2 \text{ group in an unstrained n-alkane} \right| \right)$$

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$$\cancel{\begin{matrix}\text{number of}\\ \text{carbon atoms}\\ \text{in cyclic alkane, A}\end{matrix} \times \left(\left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in the}\\ \text{cyclic alkane, A}\end{matrix} \right| - \left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in an}\\ \text{unstrained n-alkane}\end{matrix} \right| \right)}$$

$$\begin{matrix}\text{number of}\\ \text{carbon atoms}\\ \text{in cyclic alkane, A}\end{matrix} \times \left(\left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in the}\\ \text{cyclic alkane, A}\end{matrix} \right| - \right.$$

$$\left. \left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in an}\\ \text{unstrained n-alkane}\end{matrix} \right| \right)$$

$$\begin{matrix}\text{number of}\\ \text{carbon atoms}\\ \text{in cyclic alkane, A}\end{matrix}$$

$$\times \left(\left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in the}\\ \text{cyclic alkane, A}\end{matrix} \right| - \left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in an}\\ \text{unstrained n-alkane}\end{matrix} \right| \right)$$

$$\begin{matrix}\text{number of}\\ \text{carbon atoms}\\ \text{in cyclic alkane, A}\end{matrix}$$

$$\times \left(\left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in the}\\ \text{cyclic alkane, A}\end{matrix} \right| - \left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in an}\\ \text{unstrained n-alkane}\end{matrix} \right| \right)$$

$$\cancel{\begin{matrix}\text{number of}\\ \text{carbon atoms}\\ \text{in cyclic alkane}\end{matrix} \times \left(\left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in the}\\ \text{cyclic alkane}\end{matrix} \right| - \left| \begin{matrix}\text{enthalpy change of combustion}\\ \text{of a } CH_2 \text{ group in an}\\ \text{unstrained n-alkane}\end{matrix} \right| \right)}$$

(d) Given that the ΔH_f of CO_2 is -94.05 kcal mol^{-1} and ΔH_f of water is -68.3 kcal mol^{-1}, write an equation showing the formation of cyclopropane, and hence calculate the enthalpy change of formation of cyclopropane. ~~calculate the enthalpy change of formation of cyclopropane.~~ [2~~1~~]

(e) Using the formula above, prove that the ring strain energy in cyclopropane is +27.6 kcal mol^{-1}. [2]

(f) Due to the presence of ring strain, cyclopropane undergoes an addition reaction with bromine in the absence of ultraviolet radiation.

(i) Suggest the skeletal structure of the molecule formed after reaction with Br_2. [1]

(ii) Hence, using VSEPR theory, explain why the presence of ring strain causes cyclopropane to undergo addition reactions. [1]

(g) Cyclopropane rings can be formed using a technique called "cyclopropanation".

One such cyclopropanation technique involves the 2 mechanistic steps stated below:

Step 1: Dissociation of diazomethane, CH_2N_2 to form methylene, CH_2, and N_2. ~~Formation of methylene, CH_2, and N_2 from diazomethane, CH_2N_2.~~

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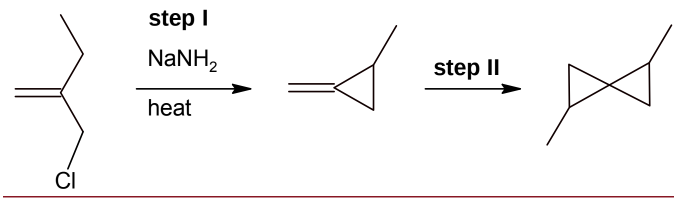
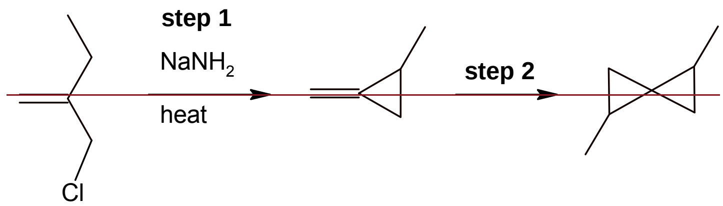
Step 2: Addition of methylene, CH_2, to trans-but-2-ene to form the cyclic ring. The reaction leaves the stereochemistry of the molecule unchanged.

(i) It is observed that the diazomethane molecule is trigonal planar in shape. By considering the shape, ~~d~~Draw a dot-and-cross diagram of diazomethane, CH_2N_2, clearly showing the type of bonds formed within the molecule. [1]

(ii) Draw the structure of the cyclic molecule formed in ~~s~~**Step 2**, showing the stereochemical arrangement clearly. State and explain if the molecule can rotate plane-polarised light. [2]

~~**(h)**~~ Cyclopropane rings are a precursor for many types of fatty acids. The following shows part of the synthetic route for fatty acids.



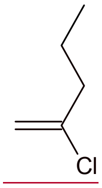


A **B**

(ii~~i~~) **Step I~~1~~** involves the reaction of molecule **A** with $NaNH_2$ to form NH_3 and a negatively-charged~~n~~ organic intermediate which eventually formed molecule **B** upon heating.

State the type of reactions that took place in **step I~~1~~** and draw the organic intermediate that was formed. [2]

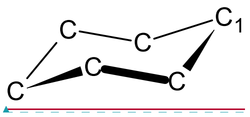
(iv~~i~~) By considering the reactivity of the Cl atom, explain why~~if~~ molecule **C** cannot be used to replace molecule **A** in the synthesis above. [1~~2~~]



C

(v) Using the above information, s~~S~~uggest a suitable reagent for **step ~~II~~2**. [1]

(h) Unlike the cyclic alkanes smaller than cyclohexane, cyclohexane does not experience ring strain due to the "chair shape" arrangement adopted by the six carbons as seen below:



chair shape arrangement of carbon atoms

By copying out the chair shape arrangement above and drawing ~~in~~ the 3-D arrangement of hydrogen atoms bonded to~~for~~ C_1, explain why cyclohexane does not experience ring strain.

[2~~2~~]

~~**(i)** Explain the reaction with $NaNH_2$. [1]~~

~~**(ii)** Hence, explain the formation of compound **C**. [1]~~

~~**(iii)** Suggest skeletal structures for compounds **B** and **C**. [2]~~

~~When a current of 1.0 A was passed through aqueous potassium maleate ($KO_2CCH{=}CHCO_2K$) for 15 minutes, it was found that 110 cm^3 H_2, measured at r.t.p, was collected at the cathode. The following reaction took place.~~

$$\cancel{2H_2O + 2e^- \rightarrow H_2 + 2OH^-}$$

~~**(d)** State the relationship between the Faraday constant, *F* and the Avogadro's constant, *L*. [1]~~

~~**(e)** Using the data above and the *Data Booklet*, calculate a value for Avogadro's constant. [3]~~

~~**(f)** Ethyne and CO_2 gas were produced at the anode. In order to determine the stoichiometry of the anode reaction, the volume of the gases collected at the anode was measured. The anode gas was first passed through aqueous NaOH before being collected in a gas syringe. The following data was collected:~~

- ~~mass of NaOH before experiment = 10.501 g~~
- ~~mass of NaOH after experiment = 10.904 g~~
- ~~initial reading on syringe = 10.0 cm^3~~
- ~~final reading on syringe = 120.0 cm^3~~

~~**(i)** State the oxidation state of carbon in ethyne. [1]~~

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~~**(ii)** With the help of an equation, explain the purpose of passing the anode gas through NaOH. [1]~~

~~**(iii)** Calculate the volume of CO_2 produced, assuming r.t.p conditions. [1]~~

~~**(iv)** Hence, suggest an ionic equation for the reaction that occurred at the anode. [1]~~

~~**(g)** When aqueous potassium maleate was acidified, maleic acid, $HO_2CCH{=}CHCO_2H$ (pK_{a1} = 1.90 and pK_{a2} = 6.07) was liberated. Fumaric acid (pK_{a1} = 3.03 and pK_{a2} = 4.44) is a stereoisomer of maleic acid.~~
~~With a suitable illustration, suggest a reason why maleic acid has a lower pK_{a1} but higher pK_{a2} than fumaric acid. [2]~~

[Total: 20]

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5(a) ~~Dopamine is an organic compound of the catecholamine and phenethylamine families that plays several important roles in the brain and body. Its name is derived from its chemical structure: it is an amine synthesised by removing a carboxyl group from a molecule of its precursor compound, L-DOPA.~~

The halogens and their compounds, show many similarities and trends in their properties. Some data are given for the elements fluorine, chlorine and iodine.

Element	Bond Energy / kJ mol^{-1}	Standard enthalpy change of atomisation / kJ mol^{-1}
Fluorine	158	79
Chlorine	242	121
Bromine	193	112
Iodine	151	107

(i) For fluorine and chlorine, their enthalpy changes of atomisation are half the value of their respective bond energies. For bromine and iodine, their enthalpy changes of atomisation are much more than half the value of their respective bond energies.

Explain in detail ~~for~~ this difference. [1]

(ii) The standard enthalpy change of formation of iodine monochloride, I–Cl, is –24.0 kJ mol^{-1}.

Use this information and the data from the table above to calculate the I–Cl bond energy. [1]

~~Below is a synthetic route involving L-DOPA and dopamine:~~

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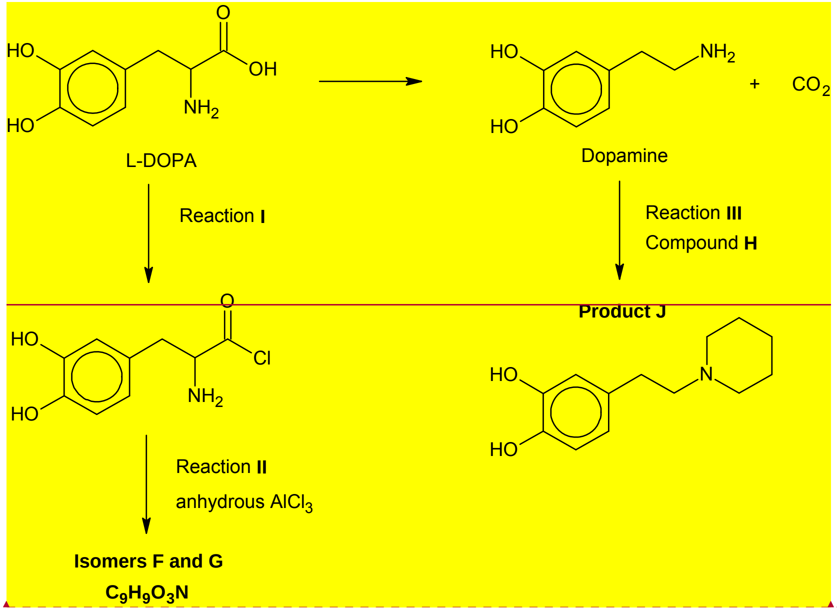
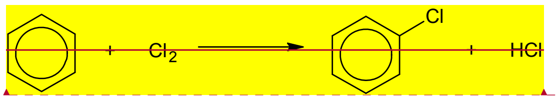


~~**(i)** State the reagents and conditions and any observations in Reaction **I**. [1]~~

~~**(ii)** Aluminium chloride is used as a catalyst in electrophilic substitution reactions. The chlorination of benzene is represented by the following overall equation.~~



~~The reaction occurs in several steps.~~

- ~~The first step is the reaction between Cl_2 and $AlCl_3$.~~

~~$Cl_2 + AlCl_3 \longrightarrow Cl^+ + AlCl_4^-$~~

- ~~The benzene ring is then attacked by the Cl^+ cation in the second step.~~

~~$AlCl_3$ reacts in a similar way with acyl chlorides, producing a carbocation that can then attack a benzene ring.~~

~~Predict the structures of isomers **F** and **G** in Reaction **II**. [2]~~

~~**(iii)** In Reaction **III**, dopamine was reacted with alkyl halide **H** to give the final product **J**. Draw the displayed formula of **H**. [2]~~

(iii) Explain why your answer in **(ii)** ~~does not correspond~~is larger in value compared to the average of the bond energies of I–I and Cl–Cl. [1]

(b) ICl reacts with pure water to form HCl and HI:

$$2ICl(l) + 2H_2O(l) \rightarrow 2HCl(aq) + 2HI(aq) + O_2(g) \qquad \Delta H_r = +171.2 \text{ kJ mol}^{-1}$$

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Using ΔH_r, the following data, as well as relevant data from **a(ii)**, draw an energy level diagram to calculate the enthalpy change of formation of aqueous HI.

Label your diagram ~~,~~ and draw arrows representing the energy terms involved. Use words or symbol to represent these energy terms.

	ΔH / kJ mol^{-1}
Standard enthalpy change of formation of H_2O	–285.8
Standard enthalpy change of formation of gaseous HCl	–92.3
Standard enthalpy change of reaction: HCl(g) $\rightarrow$ HCl(aq)	–75.1
Standard enthalpy change of vaporisation of liquid ICl	+41.4

[4]

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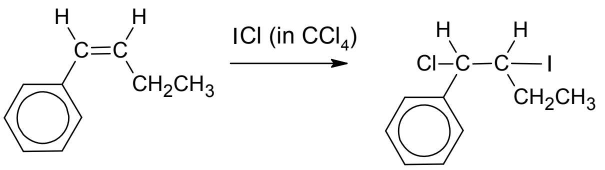
(~~b~~c) ICl is a useful reagent in organic synthesis. It is used in the following reaction to form compound **E**.

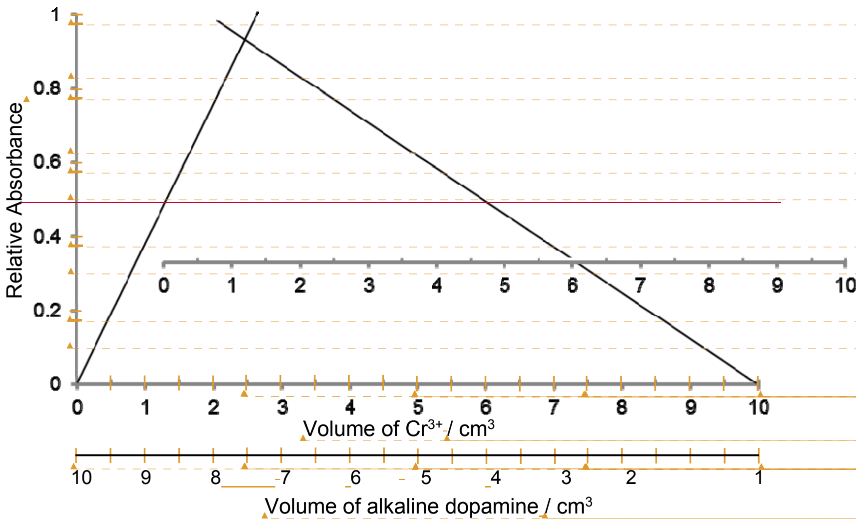


E

(i) Describe the mechanism for the formation of **E**. [3]

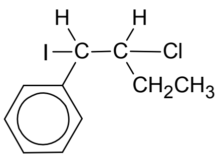
~~**(ii)** Dopamine is a bidentate ligand. When different volumes of 0.0030 mol dm^{-3} of aqueous Cr(III) and 0.0020 mol dm^{-3} of alkaline dopamine solution were mixed, a complex **R** is formed. Analysis of **R** shows that its formula is [Cr($C_8H_9NO_2$)$_x$(H_2O)$_y$]$^{z+}$, where x, y and z are integers.~~

~~To determine the stoichiometry of the complex ion formed, the colour intensities of these different mixtures were measured using a colorimeter. The following absorption spectrum was obtained.~~



~~**(i)** Use the graph and the information given to determine the formula of complex **R**. Show your workings clearly.~~ [3]

With the aid of a diagram, explain why **E** is formed and not **F**. [2]



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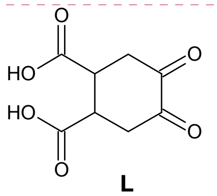
(d) Compound **J**, $C_{11}H_{16}O_2$, decolourises bromine water. 1 mole of **J** reacts with sodium metal to produce 22.7 dm^3 of hydrogen gas at s.t.p. On heating with acidified $KMnO_4$, **K**, $C_9H_{10}O_5$, is the only organic product formed.

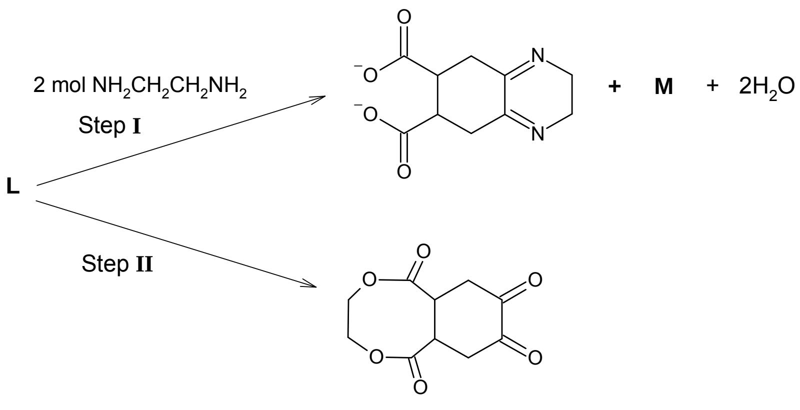
K reacts with sodium carbonate and 2,4-DNPH. When **K** reacts with alkaline aqueous iodine, **L** is formed upon acidification.



L

Suggest structures for **J** and **K** and explain the reactions described. [5]

(e) In the following reaction scheme, compounds **M** and **N** can be obtained from **L**.



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(i) Draw the structure of **M**. State the type(s) of reaction in Step **I**. [2]

(ii) Suggest reagents and conditions to synthesise product **N** from **L**. [1]

~~**(ii)** The crystal field describes the breaking of orbital degeneracy in transition metal complexes due to the presence of ligands. When the d-orbitals split into high energy and low energy orbitals, the difference in energy of the two levels is denoted as Δ_o. The relationship between Δ_o and colours of complexes can be described in the equation below:~~

$$\Delta_o = \frac{hc}{\lambda}$$

~~where h is Planck's constant, *c* is the speed of light and λ is the wavelength of light absorbed~~

~~colour~~	~~absorbed λ / nm~~
~~violet~~	~~410~~
~~indigo~~	~~430~~
~~blue~~	~~480~~
~~blue-green~~	~~500~~
~~green~~	~~530~~
~~yellow~~	~~580~~
~~orange~~	~~610~~
~~red~~	~~680~~

~~Given that Δ_o for complex **R** is 4.125 x 10^{-22} kJ and using relevant data from the *Data Booklet*, calculate the wavelength of light. Deduce the colour of complex **R**. [2]~~

~~**(c)** Iodine is not very soluble in water, it is freely soluble in KI(aq), according to the following equilibrium:~~

$$I^-(aq) + I_2(s) \rightleftharpoons I_3^-(aq)$$

~~**(i)** Draw a fully labelled experimental set-up for a voltaic cell made up of a $Cr_2O_7^{2-}$ / Cr^{3+} half-cell and a I_2/I^- half-cell under standard conditions. Indicate clearly the anode and cathode and show the flow of electrons. [3]~~

~~**(ii)** By using appropriate values from the *Data Booklet*, predict what, if anything, will happen when a small amount of acidified vanadium(II) chloride is added to the I_2/I^- half-cell. [3]~~

~~**(d)** Explain the following statements.~~

~~**(i)** BrF_3 is a covalent compound which exhibits electrical conductivity in liquid state at room temperature. With the aid of an equation, suggest an explanation for its electrical conductivity. [2]~~

~~**(ii)** $SiCl_4$ reacts violently in water but CCl_4 has no reaction with water. [1]~~

~~**(iii)** Compounds NeF_2 and NeF_4 do not exist but XeF_2 and XeF_4 exist. [1]~~

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